Course Code	**ENVIRONMENTAL ENGINEERING**	L	T	P	C
23A01603T		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the sources, demand estimation, and quality parameters of water.
2. **Apply** water treatment processes for purification and supply.
3. **Analyze** storage, distribution, and operation of water supply systems.
4. **Design** sewerage systems, stormwater drainage, and plumbing networks.
5. **Evaluate** sewage treatment, sludge management, and water reuse methods.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Explain** water sources, quality standards, and waterborne diseases.
2. **Design** unit processes of water treatment plants.
3. **Analyze** water distribution networks and pumping stations.
4. **Design** sewerage systems, including stormwater and sanitary sewers.
5. **Assess** sewage treatment methods and advanced wastewater management techniques.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	1	-	3	3	1	-	1	-	2	3	3
CO -2	3	3	3	2	2	2	2	1	-	-	-	1	3	3
CO -3	3	3	3	2	2	2	3	1	-	-	-	1	3	3
CO -4	3	3	3	2	2	2	3	1	-	-	-	1	3	3
CO -5	2	2	2	1	3	3	3	1	-	1	1	2	2	2

UNIT – I

WATER SUPPLY

Estimation of Surface and Subsurface Water Resources - Predicting Demand for Water- Impurities of Water and Their Significance - Physical, Chemical and Bacteriological Analysis - Waterborne Diseases - Standards for Potable Water. Intake of Water: Pumping and Gravity Schemes.

UNIT – II

WATER TREATMENT

Objectives - Unit Operations and Processes - Principles, Functions, and Design of Water Treatment Plant Units, Aerators of Flash Mixers, Coagulation and Flocculation – Clarifloccuator- Plate and Tube Settlers - PulsatorClarifier - Sand Filters - Disinfection - Softening, Removal of Iron and Manganese - Defluoridation- Softening - Desalination Process - Residue Management - Construction, Operation and Maintenance Aspects

UNIT – III

WATER STORAGE and DISTRIBUTION

Storage and Balancing Reservoirs - Types, Location and Capacity. Distribution System: Layout, Hydraulics of Pipe Lines, Pipe Fittings, Valves Including Check and Pressure Reducing Valves, Meters, Analysis of Distribution Systems, Leak Detection, Maintenance of Distribution Systems, Pumping Stations and Their Operations - House Service Connections.

UNIT – IV

PLANNING and DESIGN of THE SEWERAGE SYSTEM

Characteristics and Composition of Sewage - Population Equivalent - Sanitary Sewage Flow

Estimation - Sewer Materials - Hydraulics of Flow in Sanitary Sewers - Sewer Design - Storm Drainage-Storm Runoff Estimation - Sewer Appurtenances - Corrosion in Sewers - Prevention and Control – Sewage Pumping-Drainage in Buildings - Plumbing Systems for Drainage

UNIT – V

SEWAGE TREATMENT and DISPOSAL

Objectives - Selection of Treatment Methods - Principles, Functions, - Activated Sludge Process and Extended Aeration Systems - Trickling Filters - Sequencing Batch Reactor(SBR) - UASB - Waste Stabilization Ponds - Other Treatment Methods - Reclamation and Reuse of Sewage - Recent Advances in Sewage Treatment - Construction, Operation and Maintenance Aspects. - Discharge Standards-Sludge Treatment -Disposal of Sludge

TEXT BOOKS:

1. Environmental Engineering by H. S Peavy, D. R. Rowe, G. Tchobanoglous, McGraw Hill Education (India) Pvt Ltd, 2014
2. Environmental Engineering, I and II by BC Punmia, Std. Publications.

REFRENCE BOOKS:

1. Environmental Engineering, I and II by SK Garg, Khanna Publications.
2. Environmental Pollution and Control Engineering CS Rao,Wiley Publications
3. Waste water engineering by Metcalf and Eddy, McGraw Hill, 2015.
4. Environmental Engineering by D. P. Sincero and G.A Sincero, Pearson 2015.
5. Water and Waste Water Technology by Mark J Hammar and Mark J. HammarJr.Wiley, 2007.

Online Learning Resources:

https://nptel.ac.in/courses/103107084